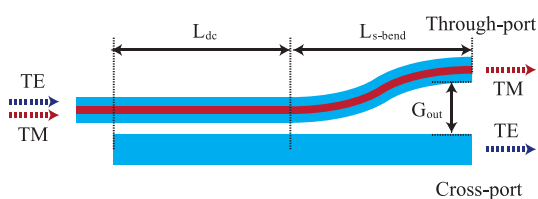


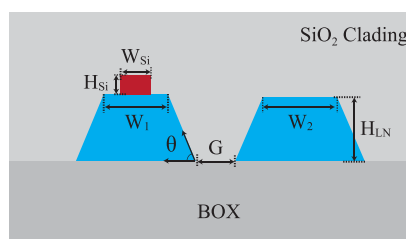
Compact, Broadband and Low-Loss Polarization Beam Splitter on Lithium-Niobate-On-Insulator Using a Silicon Nanowire Assisted Waveguide

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Compact, Broadband and Low-Loss Polarization Beam Splitter on Lithium-Niobate-On-Insulator Using a Silicon Nanowire Assisted Waveguide

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Abstract: We propose a compact, broadband, and low-loss polarization beam splitter (PBS) based on an asymmetrical directional coupler on the lithium-niobate-on-insulator platform, consisting of an amorphous silicon (a-Si) assisted waveguide, and a normal waveguide. With proper design of the a-Si nanowire, and lithium niobate waveguides, the phase-matching condition is satisfied for TE polarization, while significant mismatching exists for TM polarization. A 16- μm -long PBS (including the s-bend region) with a 200 nm gap is achieved. Numerical simulations show that the PBS exhibits a 140-nm bandwidth for extinction ratios larger than 10 dB. Moreover, the device has an ultra-low insertion loss (IL $\sim$ 0.05 dB), and maintaining ILs $<$ 0.5 dB over a 170 nm wavelength range.

Index Terms: Asymmetric directional coupler, Inoi, polarization beam splitter.

1. Introduction

Due to the high performance in electro-optic, acousto-optic, and nonlinear optic properties, lithium niobate (LN) is well known as an excellent optical material for communication science and technology. Conventional ion-diffused LN waveguides (WGs) have weak modal confinement because of the low index contrast ($\Delta n < 0.02$) between the core and cladding layer, resulting in bulky devices. Benefiting from the recent development in crystal ion slicing for thin film LN fabrication [1]–[2], lithium-niobate-on-insulator (LNOI) has emerged as a promising photonic integrated circuit (PIC) that offers sub-wavelength light confinement. Similar to the other high-index-contrast platforms, LNOI suffers from the polarization-dependent problem due to the birefringence caused by the asymmetrical waveguide structures and the inherent anisotropy property of LN [3]. Polarization diversity system [4]–[5] is a general solution to solve the polarization-dependent problem in PICs. Polarization beam splitter (PBS) is one of the most basic components in the polarization diversity

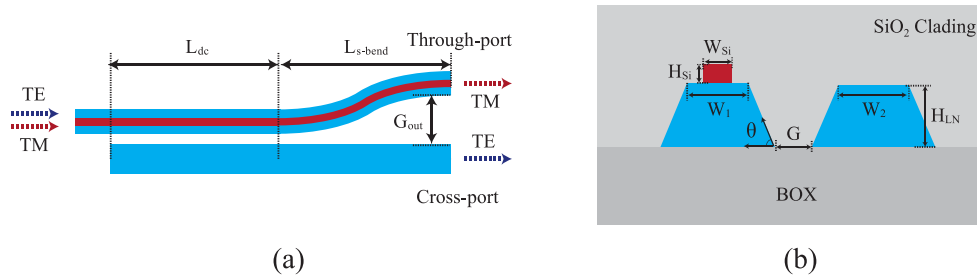


Fig. 1. Schematic of the proposed PBS. (a) top view and (b) cross-section view.

systems. Besides, PBSs are also widely used in polarization-division-multiplexing (PDM) systems [6] and quantum photonics integrated circuits [7]. However, there are lack of PBS designs on LNOI, making it difficult to control the polarizations. Recently, the adiabatic coupler [8] and Mach-Zehnder interferometer [9] have been used to realized ultra-broadband PBSs on the LNOI platform. However, these PBSs have large footprints, limiting their applications in ultra-compact PICs. A compact PBS with high performance is highly needed in LNOI-based PICs.

Asymmetrical directional coupler (ADC) has been proven to be a promising approach to realize compact PBSs with relatively simple device structures on the other relatively mature platforms, like silicon-on-insulator (SOI) [10]–[11]. It is well known that a quasi-single-mode silicon strip WG, with a typical thickness of 220 nm, exhibits ultra-high birefringence (~ 0.7). Utilizing the high birefringence, compact on-chip ADC-based PBSs can be achieved on SOI by using different widths [12]–[14]. However, those kinds of designs might not be suitable for LNOI, which has much smaller waveguide birefringence than SOI because of the higher cross-sectional aspect ratio [15]–[16]. For example, a single-mode lithium niobate (LN) strip WG with a thickness of 400 nm, exhibits birefringence typically less than 0.2. Slot WGs [17] can significantly enhance the waveguide birefringence, and small ADC-based PBS can be realized on SOI [18]–[21]. In this case, the TE-polarized coupling is sufficient while the TM-polarized coupling is efficiently suppressed. However, it is very difficult to fabricate the slot structures on the LNOI platform because of the quite small feature size with ~ 100 -nm gap width and difficult etching process of LN WGs. Besides, the sidewalls of LN WGs typically have tilt angles less than 75° , which will further increase the difficulty to fabricate the nano-slot structures. Hybrid plasmonic waveguides (HPW) [22]–[23] is another sufficient approach to significantly change the waveguide birefringence and realize compact ADC-based PBS by efficiently depressing the TE-polarized coupling [24]–[26]. However, this kind of PBS has a relatively large insertion loss because of the inherent metallic absorption.

In this paper, we propose a compact, broadband and low-loss PBS based on an asymmetrical directional coupler consisting of an amorphous silicon (a-Si) nanowire assisted LN strip WG and a normal LN strip WG. With the help of the a-Si nanowire, phase-matching condition is satisfied for TE polarization, while significant mismatching exists for TM polarization. We show for the first time that a-Si nanowires can be used to realize compact PBSs with excellent performance on the LNOI platform. Most importantly, our device is easy to fabricate due to the sample structure with large feature sizes and good fabrication tolerances.

2. Working Principle and Device Design

Fig. 1(a) and (b) show the schematic top view and cross-section view of the proposed PBS, respectively. The device consists of an a-Si nanowire assisted LN strip WG (W_1) and a normal LN strip WG (W_2) separated by a coupling gap (G). The a-Si nanowire is loaded on the top of the LN WG at the central position. The width and height of the a-Si nanowire are donated by W_{Si} and H_{Si} , respectively. For such a structure, the fabrication is CMOS compatible. The a-Si can be deposited on LNOI platform using the plasma enhanced chemical vapor (PECVD) technology

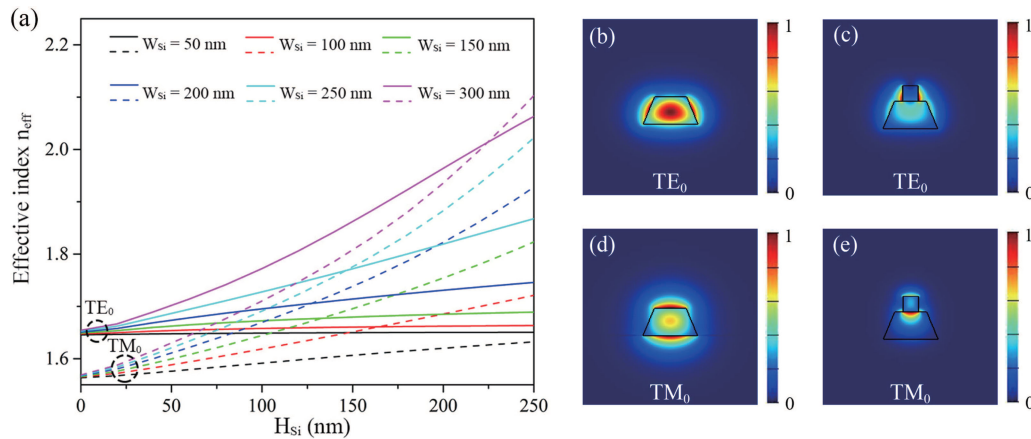


Fig. 2. (a) Calculated effective indices of TE_0 and TM_0 modes in a-Si assisted LN WGs as a function of H_{Si} for different W_{Si} . Electrical field profiles for (b) TE_0 mode (W_{Si}, H_{Si}) = (0, 0) nm, (c) TE_0 mode (W_{Si}, H_{Si}) = (200, 220) nm, (d) TM_0 mode (W_{Si}, H_{Si}) = (0, 0) nm, and (e) TM_0 mode (W_{Si}, H_{Si}) = (200, 220) nm.

[27]. Both of the LN WGs and the a-Si nanowires can be patterned using the E-beam lithography (EBL) and inductively coupled plasma reactive ion etching (ICP-RIE) to have a high resolution. Low-loss LN WGs fabricated by ICP-RIE typically have sloping sidewalls [28]–[29]. In this paper, the simulated sidewall angle (θ) is fixed at 70° to meet the state of fabrication technology. A s-bent section (L_{s-bend}) with an end separation distance (G_{out}) is connected with the straight a-Si assisted LN WG to make the two WGs decoupled. The design is based on the z-cut LNOI platform with a 400-nm LN layer (H_{LN}) and a 2- μm buried SiO_2 layer (BOX). SiO_2 is used as the upper cladding layer to protect the WGs. The refractive indices of LN, Si and SiO_2 are $n_{LN} = 2.21$, $n_{Si} = 3.48$ and $n_{\text{SiO}_2} = 1.44$, respectively.

For the proposed PBS, the a-Si nanowire geometry (W_{Si}, H_{Si}) is designed optimally to significantly increase the effective index (n_{eff}) of fundamental TM (TM_0) mode, while slightly increase n_{eff} of fundamental TE (TE_0) mode. Then with a proper design of the LN WG widths (W_1, W_2), the phase matching condition is satisfied for TE polarization. For the TM polarization with strong mode confinement, large phase mismatching exists, and the cross coupling is depressed significantly. In this case, different polarized light can be separated with a short length where the complete coupling of TE polarization occurs.

If a silicon nanowire is loaded on the LN WGs, the electric fields of TE polarization (whose main electrical field is horizontal) and TM polarization (whose main electrical field is vertical) will be influenced differently. Fig. 2(a) shows the calculated effective indices of the TE_0 and TM_0 modes in a-Si assisted LN WG at 1.55 μm wavelength. In the calculations, W_1 is fixed at 400 nm. We use the Lumericals finite element method (FEM) solver to calculate n_{eff} of the fundamental modes. It can be seen that, for both polarizations, n_{eff} increases with either of W_{Si} and H_{Si} increases. However, with the increase of H_{Si} , n_{eff} of TM_0 mode increases faster than TE_0 mode, especially for the case of smaller W_{Si} . Here, according to Fig. 2(a), we choose (W_{Si}, H_{Si}) to be (200, 220) nm as a design example to dramatically increase n_{eff} of TM_0 mode while slightly increase n_{eff} of TE_0 mode. Fig. 2(b)–(e) show the related electrical field profiles of TE_0 and TM_0 modes for cases of (W_{Si}, H_{Si}) = (0, 0) nm and (W_{Si}, H_{Si}) = (200, 220) nm. It can be seen that significant modal confinement enhancement for TM_0 mode can be observed in a-Si assisted WGs.

Then we calculate n_{eff} as a function of waveguide width at the wavelength of 1.55 μm for both kinds of WGs, as shown in Fig. 3. As expected, compared with normal LN WGs, the TM_0 modes of a-Si assisted LN WGs have significant n_{eff} increments, while TE_0 modes exhibit relatively small n_{eff} increments. Here, we choose W_1 to be 300 nm and W_2 to be 440 nm to satisfy the phase matching condition for the TE polarization. In this case, the single mode condition is satisfied for both WGs.

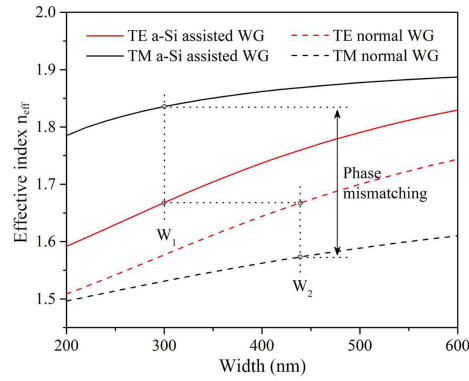


Fig. 3. Calculated effective index of TE_0 and TM_0 modes in a-Si assisted LN WGs and normal LN WGs as a function of WG width.

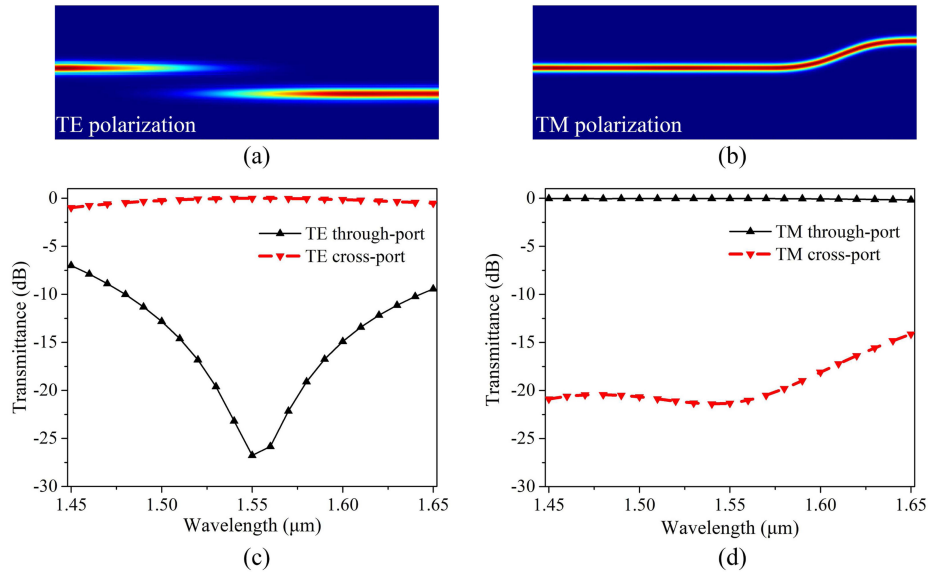


Fig. 4. Light propagation profiles of the designed PBS at $1.55 \mu\text{m}$ wavelength for (a) TE polarization and (b) TM polarization. Calculated transmittance at the through- and cross-ports for (c) TE polarization and (d) TM polarization. The design parameters are $W_{Si} = 200 \text{ nm}$, $H_{Si} = 220 \text{ nm}$, $W_1 = 300 \text{ nm}$, $W_2 = 440 \text{ nm}$, $G = 200 \text{ nm}$, $G_{out} = 1.1 \mu\text{m}$, $L_{dc} = 9.8 \mu\text{m}$, and $L_{s-bend} = 6 \mu\text{m}$.

It can be seen that, for the same polarization, the two kinds of WGs exhibit almost the same slope, which indicates large fabrication tolerance to the WG width variations.

It should be noted that the cross-over coupling length of TE_0 mode is determined by G . With a smaller G , the DC has a smaller coupling length, thus a more compact device size. However, the fabrication process decides the minimum distance between two waveguides. Here, we choose $G = 200 \text{ nm}$ for ease of fabrication. For the S-bend region, G_{out} is set to be $1.1 \mu\text{m}$ to avoid the undesired coupling between the two WGs and L_{s-bend} is designed to be $6 \mu\text{m}$ to ensure a negligible bending loss. We use the 3D finite-difference time-domain (FDTD) to simulate the light propagation and calculate the transmittance of the proposed PBS. L_{dc} is optimized to be $9.8 \mu\text{m}$ for the completed coupling of the TE_0 mode, with consideration of the s-bend coupling. Fig. 4 (a) and (b) show the calculated light propagation profiles at $1.55 \mu\text{m}$ for TE and TM polarizations, respectively. It can be seen that TE polarized light has a complete coupling to the cross-port while TM polarized light propagates without any coupling almost. The extinction ratios (ERs) are calculated to be 26.7 dB

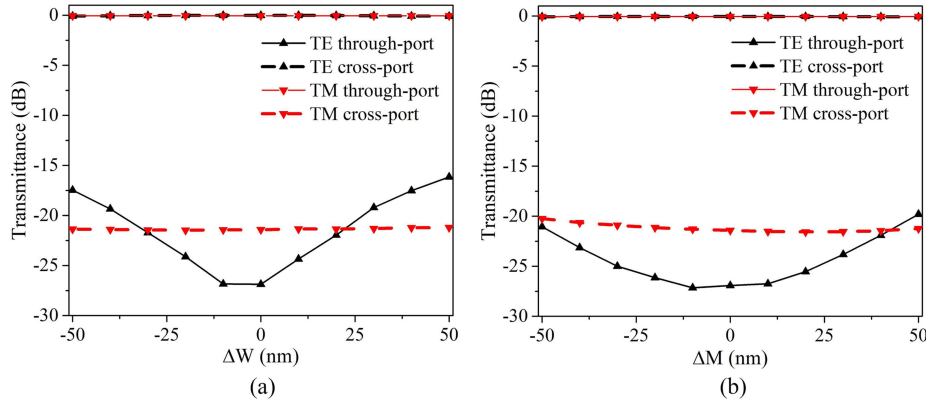


Fig. 5. Fabrication tolerance of the PBS with variational (a) ΔW and (b) ΔM .

for TE polarization and 21.3 dB for TM polarization. Ultra-low insertion losses (ILs < 0.05 dB) can be achieved for both polarizations. Here, the extinction ratio ER and insertion loss IL are defined as follows,

$$ER_{TE} = 10 \log_{10}(T_{cross-port, TE} / T_{through-port, TE}) \quad (1)$$

$$ER_{TM} = 10 \log_{10}(T_{through-port, TM} / T_{cross-port, TM}) \quad (2)$$

$$IL_{TE} = 10 \log_{10}(T_{cross-port, TE}) \quad (3)$$

$$IL_{TM} = 10 \log_{10}(T_{through-port, TM}) \quad (4)$$

Where $T_{through-port}$ and $T_{cross-port}$ are the transmittances at the through-port and cross-port, respectively. Fig. 4(c) and (d) show the simulated transmittance spectra for TE and TM polarizations, respectively. It can be observed that the PBS exhibits broadband performance with bandwidth of 140 nm for ERs > 10 dB for both polarizations. The IL is < 0.5 dB over a 150 nm bandwidth for TE polarization and < 0.2 dB for TM polarization over a 200 nm bandwidth. The ILs for TM polarization mainly result from the bend-induced losses and can be further reduced by increasing L_{s-bend} .

Fabrication error tolerance of the proposed device is also analyzed. Fig. 5(a) and (b) show the calculated transmittance as a function of waveguide width variation (ΔW) and mask misalignment variation (ΔM). We assume that the two adjacent waveguides, W_1 and W_2 , have the same variation ΔW , while the gap, G , has an opposite variation, $\Delta G = -\Delta W$. ΔM is defined as the horizontal offset of the silicon nanowire relative to the normal LN WG. It can be observed that the PBS exhibits large tolerance to the fabrication error maintaining ERs > 16 dB for ΔW and ERs > 20 dB for ΔM within 100-nm variation range for both polarizations. The influence of the misalignment is smaller than the waveguide variation. This large fabrication tolerance ensures our proposed PBS easy to fabricate, even though an overlay exposure is needed. It should be noted that TM polarization has much lower sensitivity to both of fabrication error and wavelength variations than TE polarization, which result from the large phase mismatching of TM_0 mode.

3. Conclusion

In this paper, we have proposed a compact, broadband and low-loss PBS based on an asymmetrical directional coupler on the LNOI platform. An amorphous silicon nanowire is introduced to the LN waveguide to significantly change the birefringence. With proper design of the a-Si nanowires and LN WGs, the phase-matching condition is satisfied for TE polarization. However, for the TM polarization, there is significant phase mismatching and the coupling is sufficiently depressed. A 16- μm -long PBS (including the s-bend region) with 200-nm coupler gap has been designed as an example. The numerical simulations show that the PBS has a broadband performance maintaining ERs > 10 dB for a 140-nm bandwidth. Moreover, the PBS exhibits an ultra-low IL (~ 0.05 dB) and

maintaining ILs < 0.5 dB over a 170 nm wavelength range. The device is easy to fabricate because of the relatively large feature size of 200 nm and good tolerance to the fabrication errors. We believe our works pave ways for the integrated polarization diversity/(de-)multiplexing systems on the LNOI platform.

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